Information Security Project

1 Objective

Modern websites and applications can track users using many mechanisms beyond traditional cookies, including browser and device fingerprinting, storage mechanisms, identifiers, behavioral signals, cross-context information, and combinations of multiple tracking techniques.

In this project, each team will investigate advanced tracking from either an **offensive or defensive Information Security perspective**. The goal is to approach the project as a small research study: understand existing work, identify an interesting problem or limitation, design and implement a technically meaningful approach, and evaluate it experimentally.

2 Research and Related Work

Each team must study **at least one scientific paper** related to its chosen direction. The papers provided below are starting references, but students are strongly encouraged to search for additional and more recent work.

The related-work study should go beyond summarizing a paper. Students should understand the proposed technique, methodology, and results, discuss its limitations, and explain how their own project relates to or goes beyond it.

The proposed papers are **references, not specifications**. Students have substantial freedom to reproduce, combine, improve, challenge, or extend existing techniques, and are highly encouraged to explore new ideas.

3 Project Directions

3.1 Offensive Track — Advanced Tracking

Offensive teams should design and implement **advanced tracking techniques** with the objective of making users, browsers, or devices as distinguishable and trackable as possible within the controlled experimental environment.

The system should aim for high **uniqueness, stability, persistence, robustness, and efficiency**. Teams may reimplement or improve existing tracking techniques, combine several mechanisms, study how multiple weak signals become powerful when combined, or investigate how tracking remains possible when some mechanisms are blocked or modified.

Students are particularly encouraged to think beyond well-known techniques and investigate **new signals, features, correlations, or combinations of information** that could improve tracking. Novelty does not necessarily require

inventing an entirely new mechanism: a new combination, improved measurement methodology, or demonstration that an existing signal works under previously unexplored conditions can also constitute an interesting contribution.

The goal is not simply to collect standard browser attributes, but to ask:

> *How far can tracking be pushed, and what additional information can make identification more reliable or persistent?*

Suggested starting reference:

S. Bouhenniche, P. Laperdrix, and W. Rudametkin, *"EXADPrinter: Semi-Exhaustive Permissionless Device Fingerprinting Within the Android Ecosystem,"* Proceedings on Privacy Enhancing Technologies (PETS), 2026.

3.2 Defensive Track — Detection, Transparency and Evaluation

Defensive teams may investigate how advanced tracking can be **detected, measured, understood, exposed, compared, or prevented**.

One possible direction is a **large-scale measurement or evaluation study**. Teams may analyze websites, applications, or tracking scripts to identify which tracking mechanisms are used, compare several detection or privacy-protection techniques, or evaluate how well existing defenses resist advanced tracking mechanisms.

Another direction is to build a **detection or transparency tool**, such as a browser extension, or dashboard. Such a system could detect tracking activity, identify the techniques being used, show which scripts or third parties are involved, and provide users or researchers with an understandable view of otherwise invisible tracking behavior.

Projects should go beyond detecting only simple cookies or obvious API calls. Teams should consider **advanced, combined, or less visible tracking mechanisms**, and where appropriate evaluate their approach at meaningful scale.

Suggested starting reference:

U. Iqbal, S. Englehardt, and Z. Shafiq, *"Fingerprinting the Fingerprinters: Learning to Detect Browser Fingerprinting Behaviors,"* IEEE Symposium on Security and Privacy, 2021.

3.3 Defensive Track — Protection

Teams may also focus directly on **protecting users against advanced tracking**.

Possible approaches include randomization, spoofing, standardization, blocking, isolation, controlled noise, modification of exposed information, or combinations of several mechanisms.

Projects may either propose a new protection mechanism or perform a rigorous comparison and evaluation of existing defenses against advanced tracking techniques.

A good protection project should consider both sides of the problem:

> *How much tracking is prevented, and what does the protection cost in terms of compatibility, performance, or usability?*

Students should also think adversarially: if a protection modifies one tracking signal, can the tracker simply rely on another one?

Suggested starting reference:

P. Laperdrix, B. Baudry, and V. Mishra, *"FPRandom: Randomizing Core Browser Objects to Break Advanced Device Fingerprinting Techniques,"* ESSoS, 2017.

4 Experimental Evaluation

A working implementation alone is not sufficient. Each team must formulate a **clear research question** and answer it through experiments.

The evaluation should use appropriate baselines and metrics. Depending on the project, these may include uniqueness, persistence, re-identification rate, detection precision and recall, tracking coverage, robustness against defenses, evasion rate, performance overhead, website compatibility, or scalability.

Negative results are also valuable when supported by a sound methodology and careful analysis.

5 Ethical Requirements

All experiments must be performed on controlled systems, authorized infrastructure, test profiles, public datasets where appropriate, or consenting participants. Tracking mechanisms must not be deployed against unsuspecting users, and unnecessary personal or sensitive data must not be collected.

Within these boundaries, students are encouraged to be ambitious and creative.

6 Final Paper and Presentation

The final results must be presented as a **short and concise scientific paper**, followed by an end-of-semester presentation.

The paper should follow the **ACM AsiaCCS formatting and submission style**:

`https://asiaccs2027.cityu.edu.mo/call-for-papers/index.html`

The official AsiaCCS page limit does **not** apply to the course project. The paper should instead be concise while clearly presenting the problem, related work, approach, experimental methodology, results, limitations, and conclusions.

7 Potential Research Extension

Projects showing particularly **innovative ideas, strong implementations, or interesting experimental findings** may have potential beyond the course. Interested students can contact me, and we will decide whether the work could be extended and developed toward a real research paper or conference/workshop submission.